On the Structural Impossibility of Incremental Concept Learning in Dense Weight Architectures

A Formal Analysis via Spectral Perturbation Theory

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Abstract

We prove that safe incremental learning of structurally new concepts is architecturally impossible in dense weight neural networks—not for algorithmic reasons, but for structural ones rooted in spectral perturbation theory.

The core result (Theorem 2) shows that in any associative memory with symmetric weight matrix W, no perturbation ΔW—of any rank or produced by any optimisation method—can introduce a new attractor a^* without necessarily changing the eigenstructure of W: either existing eigenvectors are rotated or existing eigenvalues shift. In either case, existing stored attractors are at risk. The result is purely structural: it follows from the symmetry of ΔW and the completeness of the eigenbasis, independent of any specific learning algorithm.

For Transformer architectures, we prove the Connectome Equivalence Lemma (Lemma 8): any parameter change $(\Delta K, \Delta V)$ that introduces a new attractor while preserving all existing ones must solve a linear system whose coefficients are the existing attractor positions themselves. Safe incremental learning is therefore structurally equivalent to knowing the implicit connectome—the full set of semantic attractors encoded in the weight matrices—which is not observable from the network parameters alone. This is formally analogous to the Knill–Laflamme conditions in quantum error correction: correcting errors requires knowing the code.

We further establish: a spectral survival condition quantifying when existing attractors survive parameter changes (Proposition 5); an explicit softmax-competition formula for the appended key-value case (Lemma 7); and the minimal architectural conditions for non-destructive concept addition (Proposition 9).

Catastrophic forgetting, in its deepest form, is not an engineering deficiency. It is a structural consequence of implicit topology.

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1 Introduction

The problem of catastrophic forgetting—the tendency of neural networks to lose previously learned knowledge when trained on new data—has been studied for over three decades [6, 8]. The standard framing treats it as an engineering problem: the network overwrites old weights when exposed to new data, and the remedy is better regularisation, rehearsal, or architectural isolation.

We argue that this framing is incorrect for an important class of learning problems. There is a categorical distinction between:

1. *New facts about existing concepts*: associating a known concept with new information. This requires only weight modification within the existing eigenspace and is, in principle, tractable.
2. *Structurally new concepts*: concepts whose attractor structure does not yet exist in the eigenspace of the weight matrix. Introducing such an attractor provably requires global reorganisation—no local operation suffices.

The second case is not an engineering problem. It is a structural one. No algorithm can solve a structural problem.

This paper provides a formal proof of this claim for associative memory networks and a precisely stated conjecture for Transformer architectures. The broader wave-dynamic framework from which this distinction emerged is developed in [1].

2 Background and Related Work

2.1 Hopfield Networks and Associative Memory

The classical Hopfield network [3] stores patterns $\xi^1, \ldots, \xi^p \in \{-1, +1\}^n$ as attractors of the energy function $E(s) = -\frac{1}{2} s^\top W s$, where the weight matrix is constructed by Hebbian learning: $W = \frac{1}{n} \sum_k \xi^k (\xi^k)^\top$.

The modern Hopfield network [7] extends this to continuous states with exponential storage capacity, and—crucially for our purposes— establishes that Transformer self-attention is mathematically equivalent to one step of the modern Hopfield network update rule. This equivalence is the bridge between our results for associative memory and the conjecture for Transformers.

2.2 Catastrophic Forgetting

McCloskey and Cohen [6] identified catastrophic interference as a fundamental problem in connectionist models. Subsequent work has proposed many mitigation strategies: Elastic Weight Consolidation (EWC) [5], Progressive Neural Networks [9], and low-rank adaptation methods such as LoRA [4].

None of these approaches address the structural question: *can any weight change—local or global—introduce a qualitatively new attractor without disturbing existing ones, even in principle?* We show the answer is no for associative memory, and that for Transformers, any safe such change requires knowing the full set of existing attractors (Lemma 8).

The present work differs from the continual learning literature in that we do not propose a new algorithm or regularisation scheme. We prove a structural lower bound: no algorithm can guarantee safe concept addition in a dense weight architecture without access to the implicit connectome. This reframes catastrophic forgetting as an information-theoretic problem, not an optimisation one.

2.3 Spectral Perturbation Theory

The Davis-Kahan theorem [2] provides quantitative bounds on eigenvector perturbations: for symmetric matrices W and perturbation ΔW,

$$\sin \angle(e_i, \tilde{e}_i) \leq \frac{\|\Delta W\|_2}{\delta_i},$$

where $\delta_i = \min_{j \neq i} |\lambda_i - \lambda_j|$ is the spectral gap. This bound shows that any *non-zero* perturbation can rotate the i-th eigenvector, and the rotation grows with $\|\Delta W\|_2$. Our contribution goes further: we show that for introducing a new attractor, the shift is not merely possible but structurally necessary—it cannot be avoided by any choice of ΔW.

3 Formal Framework

Definition 1 (Hopfield Network)**.** *Let $W \in \mathbb{R}^{n \times n}$ be symmetric with $W_{ii} = 0$. The energy function is $E(s) = -\frac{1}{2} s^\top W s$ for $s \in \{-1, +1\}^n$. The asynchronous update rule is $s_i \leftarrow \operatorname{sign}\left(\sum_j W_{ij} s_j\right)$.*
Definition 2 (Attractor)**.** *A pattern $\xi \in \{-1, +1\}^n$ is an attractor of W if and only if $\operatorname{sign}(W\xi) = \xi$, i.e., ξ is a fixed point of the update rule.*
Definition 3 (New Concept)**.** *A vector $a^* \in \mathbb{R}^n$ is a* structurally new concept *with respect to trained network W if $\langle a^*, \xi^k \rangle \approx 0$ for all stored attractors $\xi^1, \ldots, \xi^p$. That is, a^* lies outside the span of the stored patterns.*

4 Main Results

4.1 Attractor Approximation (Lemma)

Lemma 1 (Attractors Approximate Eigenvectors)**.** *Let $W = \frac{1}{n} \sum_{k=1}^{p} \xi^k (\xi^k)^\top$ be the Hebbian weight matrix for patterns $\xi^1, \ldots, \xi^p$. Then for each stored pattern ξ^m:*

$$W\xi^m = \xi^m + \frac{1}{n} \sum_{k \neq m} \langle \xi^k, \xi^m \rangle \xi^k.$$

When $p \ll n$ and patterns are approximately orthogonal, the cross-talk term is small, and ξ^m is an approximate eigenvector of W with eigenvalue ≈ 1.

Proof. Direct computation: $W\xi^m = \frac{1}{n} \sum_k \xi^k (\xi^k)^\top \xi^m = \frac{1}{n} \sum_k \langle \xi^k, \xi^m \rangle \xi^k = \xi^m + \frac{1}{n} \sum_{k \neq m} \langle \xi^k, \xi^m \rangle \xi^k$, where we used $\langle \xi^m, \xi^m \rangle = n$. When $p \ll n$, the patterns are approximately orthogonal by a counting argument, and the residual term has norm $O(p/\sqrt{n})$, which vanishes as $n \to \infty$ at fixed p [3]. □

4.2 The Impossibility Theorem

Theorem 2 (New Concept Requires Global Eigenstructure Change)**.** *Let $W \in \mathbb{R}^{n \times n}$ be symmetric with eigendecomposition $W = \sum_i \lambda_i e_i e_i^\top$, where $\{e_i\}$ is an orthonormal basis of $\mathbb{R}^n$. Let $a^* \in \mathbb{R}^n$ be a structurally new concept with $\langle a^*, e_i \rangle \approx 0$ for $i = 1, \ldots, p$ (orthogonal to all stored attractors), and assume a^* is not already an eigenvector of W (i.e., $Wa^* \neq \mu a^*$ for any μ).*

Then for any symmetric perturbation ΔW that makes a^* an attractor of $W' = W + \Delta W$ (i.e. $W'a^* = \mu a^*$ for some $\mu > 0$), the eigenstructure of W' necessarily changes relative to W:

1. *at least one existing eigenvector e_j is no longer an eigenvector of W' (eigenvector rotation), or*
2. *at least one eigenvalue shifts ($\lambda'_j \neq \lambda_j$ for some j).*

The approximate case $W'a^* \approx \mu a^*$ follows by continuity of eigenvalues and eigenvectors under small perturbations (Weyl's inequality). In both cases the eigenstructure of W' differs from that of W; the quantitative conditions under which this disrupts existing stored attractors are given in Proposition 5.

Proof. We work in the exact regime $W'a^* = \mu a^*$; the approximate case follows by continuity of eigenvalues and eigenvectors (Weyl's inequality). The proof splits according to whether the eigenvectors of W are preserved as eigenvectors of W'.

Case 1: Some eigenvector e_j is no longer an eigenvector of W'.

If $W'e_j \neq \lambda' e_j$ for any λ', then e_j has been rotated in W'. The eigenstructure of W' differs from that of W; case (a) holds.

Case 2: All $e_1, \ldots, e_n$ remain eigenvectors of W' (eigenvalues must shift).

Suppose $W'e_j = (\lambda_j + c_j)e_j$ for all j, so $\Delta W \cdot e_j = c_j e_j$. We show that at least one $c_j \neq 0$.

Step A: Residual vector. Since $W'a^* = \mu a^*$:

$$\Delta W \cdot a^* = \mu a^* - Wa^* = \sum_i (\mu - \lambda_i)\langle e_i, a^*\rangle e_i =: r.$$

Since a^* is not an eigenvector of W, at least one summand $(\mu - \lambda_j)\langle e_j, a^*\rangle \neq 0$, so $r \neq 0$.

Step B: Compatibility via symmetry. For any j, compute $\langle e_j, \Delta W \cdot a^*\rangle$ two ways: from Step A it equals $(\mu - \lambda_j)\langle e_j, a^*\rangle$; from the Case 2 assumption and symmetry of ΔW it equals $\langle \Delta W e_j, a^*\rangle = c_j\langle e_j, a^*\rangle$. Equating:

$$c_j\langle e_j, a^*\rangle = (\mu - \lambda_j)\langle e_j, a^*\rangle \quad \text{for all } j. \tag{$*$}$$

Step C: Eigenvalue shift. Since $a^* \neq 0$ and $\{e_j\}$ is a complete basis, there exists j with $\langle e_j, a^*\rangle \neq 0$. For any such j, equation $(*)$ gives $c_j = \mu - \lambda_j$. If $c_j \neq 0$ for any such j, the eigenvalue of e_j shifts: $\lambda'_j = \lambda_j + c_j = \mu$. It remains to rule out $c_j = 0$ for every j with $\langle e_j, a^*\rangle \neq 0$. This would require $\lambda_j = \mu$ for all such j, meaning every non-zero component of a^* lies in the μ-eigenspace of W, i.e. $Wa^* = \mu a^*$—contradicting the hypothesis that a^* is not already an eigenvector of W. Therefore $c_j \neq 0$ for some j: at least one eigenvalue shifts.

In both cases the eigenstructure of W' necessarily differs from that of W. This holds for any symmetric ΔW, regardless of rank or optimisation method. □ □

Remark 1 (From Eigenstructure Change to Attractor Disruption)**.** *Theorem 2 establishes a* necessary *structural change in W': either eigenvectors rotate (case a) or eigenvalues shift (case b). This does not by itself guarantee that a stored attractor is destroyed—a small perturbation may rotate an eigenvector by very little. The quantitative threshold is given by Proposition 5: attractor ξ^m is at risk whenever the required eigenvalue change $|\mu - \lambda_m|$ meets or exceeds the spectral gap δ_m. Thus the full impossibility argument has two parts: (i) any introduction of a^* forces an eigenstructure change (Theorem 2); (ii) that change disrupts ξ^m whenever $|\mu - \lambda_m| \geq \delta_m$ (Proposition 5).*

Corollary 3 (No Algorithm Circumvents the Result)**.** *Theorem 2 applies to rank-1, rank-k, and full-rank perturbations. For asymmetric methods such as LoRA ($\Delta W = BA^\top$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times d}$), note that the effective weight change in a symmetric Hopfield network is the symmetrised form $\frac{1}{2}(BA^\top + AB^\top)$, to which the theorem applies directly. More generally, the proof requires only that ΔW acts on $\mathbb{R}^n$ with a complete eigenbasis; the symmetry plays a role*

only in Step C. Any training procedure that modifies a real weight matrix—symmetric or not—cannot avoid an eigenstructure change when introducing a genuinely new attractor. The result is structural, not algorithmic: no choice of learning rate, regulariser, or optimiser can avoid it.

Proof. The proof of Theorem 2 uses symmetry in Step C to exchange $\langle e_j, \Delta W \cdot a^* \rangle = \langle \Delta W^\top e_j, a^* \rangle$. For symmetric ΔW this equals $\langle \Delta W \cdot e_j, a^* \rangle = c_j \langle e_j, a^* \rangle$. For asymmetric perturbations, the same argument applies to the symmetrised part $\frac{1}{2}(\Delta W + \Delta W^\top)$, which governs the eigenvalue changes of the symmetric update $W' = W + \frac{1}{2}(\Delta W + \Delta W^\top)$. □

4.3 Distinction Between Two Learning Cases

Theorem 2 immediately yields a categorical distinction:

Corollary 4 (Two Qualitatively Different Learning Cases)**.** *1. **New facts about existing concepts**: If the new information can be encoded as a weight shift within the current eigenspace (i.e., modifications to existing attractors), then a^* is already in* $\mathrm{span}\{e_i\}$*, and the learning is tractable in principle. Catastrophic forgetting remains a risk due to the global nature of ΔW, but is not structurally necessary.*

*2. **Structurally new concepts**: If the new concept requires an attractor a^* orthogonal to all existing eigenvectors, then by Theorem 2, its introduction necessarily disrupts the eigenstructure of W: either existing eigenvectors are rotated (displacing stored attractors from their basins) or at least one eigenvalue shifts. By Proposition 5, this puts at least one existing attractor at risk. This case is structurally unavoidable without full retraining.*

4.4 Quantitative Bounds

The proof of Theorem 2 yields two further results that resolve Open Problems 1 and 3, and a displacement bound providing partial progress on Open Problem 2.

Proposition 5 (Survival Condition for Existing Attractors)**.** *Under the conditions of Theorem 2, let $\delta_m = \min_{j \neq m} |\lambda_m - \lambda_j|$ be the spectral gap of attractor ξ^m. A necessary condition for ξ^m to survive the introduction of new concept a^* with target eigenvalue μ is:*

$$|\mu - \lambda_m| < \delta_m.$$

If this condition is violated, any perturbation ΔW introducing a^* has $\|\Delta W\|_2 \geq |\mu - \lambda_m| \geq \delta_m$, beyond the Davis-Kahan safe radius for e_m; preservation of ξ^m is not guaranteed.

Proof. From Step D of Theorem 2, any symmetric ΔW introducing a^* must satisfy $\Delta W \cdot e_m = c_m e_m$ with $c_m = \mu - \lambda_m$ (for $\langle e_m, a^* \rangle \neq 0$), giving $\|\Delta W\|_2 \geq |c_m| = |\mu - \lambda_m|$. For $\xi^m \approx e_m$ to remain a stable attractor of W', a sufficient condition is that the eigenspace of W' near λ_m remains close to e_m. By the Davis-Kahan theorem [2], this holds when $\|\Delta W\|_2 < \delta_m$. The condition $|\mu - \lambda_m| < \delta_m$ is therefore necessary for this sufficient condition to be achievable. □

Remark 2. *Proposition 5 predicts that concepts with dense semantic neighbourhoods (small δ_m) are most vulnerable to catastrophic forgetting: any new concept whose target eigenvalue is farther than δ_m from λ_m risks destroying ξ^m. Concepts in sparse semantic regions (large δ_m) are robust. This is consistent with empirical observations that fine-tuning disrupts common, densely-represented concepts more than rare ones.*

Proposition 6 (Geometric Threshold: Case 1 vs. Case 2 (Heuristic))**.** *Let $P_p = \sum_{i=1}^{p} e_i e_i^\top$ be the orthogonal projection onto the eigenspace of the p stored concepts, and define the* novelty angle*:*

$$\cos\theta(a^*) = \frac{\|P_p\, a^*\|}{\|a^*\|} \in [0, 1].$$

This proposition is heuristic; a formal proof is an open problem. *The natural operational threshold separating Case 1 from Case 2 is* $\cos\theta(a^*) = 1/\sqrt{p}$*:*

- $\cos\theta > 1/\sqrt{p}$*: the projection of* a^* *onto the existing eigenspace exceeds the expected cross-talk of a random unit vector (*$O(1/\sqrt{p})$ *by a counting argument [3]). The new concept is substantially representable within the current eigenspace:* ***Case 1****.*
- $\cos\theta < 1/\sqrt{p}$*:* a^* *lies effectively outside the current eigenspace. By Corollary 4, its introduction requires an eigenstructure change:* ***Case 2****.*

Remark 3 (Displacement Bound via the Implicit Function Theorem)**.** *The displacement of an existing Transformer attractor* c_C *following a parameter change* $\Delta\theta$ *can be bounded via the implicit function theorem. Let* $g = f_{\theta+\Delta\theta} - f_\theta$ *and* $J = \partial f_\theta/\partial x|_{c_C}$*. Since* $\rho(J) < 1$*,* $(I - J)$ *is invertible and the displaced fixed point satisfies:*

$$\|c'_C - c_C\| \leq \frac{\|g(c_C)\|}{1 - \rho(J)}.$$

The remaining open step is to show that $g(c_C) \neq 0$ *whenever a genuinely new attractor* c_{a^*} *requires incompatible Jacobi structure—i.e., that existing fixed points cannot be preserved when the linearised system must accommodate a new dominant singular direction. This is the essential form of Conjecture 1.*

Lemma 7 (Softmax Competition: Open Problem 2 Resolved)**.** *Let* c_C *be a semantic attractor of attention* f_θ *with* $f_\theta(c_C) = c_C = \sum_{i=1}^{p} \alpha_i(c_C)\, v_i$*, where* $\alpha_i(c_C) = \mathrm{softmax}_i(c_C K^\top/\sqrt{d})$*. Introduce a new concept* a^* *by appending key* k_{a^*} *and value* v_{a^*} *to* K *and* V *respectively, yielding modified attention* $f_{\theta'}$*. Write:*

$$\alpha'_{a^*}(c_C) = \frac{\exp(c_C k_{a^*}^\top/\sqrt{d})}{\displaystyle\sum_{j=1}^{p} \exp(c_C k_j^\top/\sqrt{d}) + \exp(c_C k_{a^*}^\top/\sqrt{d})}.$$

Then:

$$g(c_C) \;=\; f_{\theta'}(c_C) - c_C \;=\; \alpha'_{a^*}(c_C)\,\big(v_{a^*} - c_C\big).$$

In particular, $g(c_C) \neq 0$ *whenever* $\alpha'_{a^*}(c_C) > 0$ *and* $v_{a^*} \neq c_C$*. Combined with Remark 3, this gives the explicit displacement bound:*

$$\|c'_C - c_C\| \;\leq\; \frac{\alpha'_{a^*}(c_C)\,\|v_{a^*} - c_C\|}{1 - \rho(J)}.$$

Proof. Adding k_{a^*} rescales the existing softmax weights uniformly: $\alpha'_i(c_C) = (1 - \alpha'_{a^*}(c_C))\,\alpha_i(c_C)$ for $i = 1, \ldots, p$. Therefore:

$$\begin{aligned} f_{\theta'}(c_C) &= \sum_{i=1}^{p} \alpha'_i(c_C)\, v_i + \alpha'_{a^*}(c_C)\, v_{a^*} \\ &= \big(1 - \alpha'_{a^*}(c_C)\big) \sum_{i=1}^{p} \alpha_i(c_C)\, v_i + \alpha'_{a^*}(c_C)\, v_{a^*} \\ &= \big(1 - \alpha'_{a^*}(c_C)\big)\, c_C + \alpha'_{a^*}(c_C)\, v_{a^*}, \end{aligned}$$

where the last step uses the fixed-point condition $\sum_i \alpha_i(c_C)\, v_i = c_C$. Subtracting c_C yields the result. □

Remark 4. *The condition* $v_{a^*} \neq c_C$ *is satisfied whenever* a^* *is a genuinely new concept with a distinct value representation. The condition* $\alpha'_{a^*}(c_C) > 0$ *holds unless* k_{a^*} *is explicitly designed to be orthogonal to all existing attractors—the construction used in Proposition 9 below. Lemma 7 thus resolves Open Problem 2 for the class of parameter changes that append new key-value pairs: the displacement is non-zero and explicitly quantified.*

5 Extension to Transformer Architectures

5.1 Transformers as Modern Hopfield Networks

Ramsauer et al. [7] proved that the Transformer self-attention operation is equivalent to one update step of a modern Hopfield network with energy function:

$$F(x) = -\mathrm{lse}(xK^\top/\sqrt{d}) + \tfrac{1}{2}x^\top x + C,$$

where lse denotes the log-sum-exp function. The attention fixed point satisfies:

$$x^* = \mathrm{softmax}(x^* K^\top/\sqrt{d}) \cdot V. \qquad \text{(Self-consistency)}$$

Definition 4 (Semantic Attractor in a Transformer)**.** *Let $R \in \mathbb{R}^d$ be the residual stream at layer l for token t. We say R represents concept C if $\|R - c_C\| < \varepsilon$ for a stable vector $c_C \in \mathbb{R}^d$ satisfying the self-consistency condition above. c_C is the* semantic attractor *of concept C at layer l.*

5.2 Structural Conjecture

The Jacobi matrix of the attention fixed-point iteration at attractor c_C is:

$$J = \left.\frac{\partial f}{\partial x}\right|_{x=c_C}, \quad f(x) = \mathrm{softmax}(xK^\top/\sqrt{d}) \cdot V.$$

Stability of c_C requires $\rho(J) < 1$ (spectral radius). Semantic attractors correspond to dominant directions of J.

Lemma 8 (Connectome Equivalence: Preservation Requires Knowing All Attractors)**.** *Let $f_\theta(x) = \mathrm{softmax}(xK^\top/\sqrt{d}) \cdot V$ be single-head attention with existing attractors $c_C^{(1)}, \ldots, c_C^{(p)}$. Let $(\Delta K, \Delta V)$ be any parameter change satisfying:*

1. Introduction*: $f_{\theta'}(a^*) = a^*$ for some new concept a^*;*
2. Preservation*: $f_{\theta'}(c_C^{(m)}) = c_C^{(m)}$ for all $m = 1, \ldots, p$.*

Then ΔV must satisfy the linear system

$$\sum_{j=1}^{p} \alpha'_j(c_C^{(m)})\, \Delta v_j \;=\; r^{(m)} \quad \textit{for all } m = 1, \ldots, p, \qquad (\star)$$

where $\alpha'_j(c_C^{(m)}) = \mathrm{softmax}_j(c_C^{(m)} K'^\top/\sqrt{d})$ and $r^{(m)} = c_C^{(m)} - \sum_j \alpha'_j(c_C^{(m)})\, v_j$.

Both the coefficient matrix $A_{mj} = \alpha'_j(c_C^{(m)})$ and the right-hand side $r^{(m)}$ depend explicitly *on the attractor positions $c_C^{(1)}, \ldots, c_C^{(p)}$. Therefore:*

Any algorithm computing a valid $(\Delta K, \Delta V)$ must have access
to all existing attractor positions—i.e., to the full implicit connectome.

Proof. Condition (ii) expanded gives, for each m:

$$\sum_{j=1}^{p} \alpha'_j(c_C^{(m)})(v_j + \Delta v_j) \;=\; c_C^{(m)}.$$

Rearranging and using the original fixed-point structure:

$$\sum_j \alpha'_j(c_C^{(m)})\, \Delta v_j \;=\; c_C^{(m)} - \sum_j \alpha'_j(c_C^{(m)})\, v_j =: r^{(m)},$$

which is exactly system ($\star$). Since $K' = K + \Delta K$, the weights $\alpha'_j(c_C^{(m)}) = \mathrm{softmax}_j(c_C^{(m)}(K + \Delta K)^\top/\sqrt{d})$ depend on $c_C^{(m)}$ and on the chosen ΔK. Similarly, $r^{(m)}$ contains $c_C^{(m)}$ on its right-hand side. No computation of A_{mj} or $r^{(m)}$ is possible without explicit knowledge of $c_C^{(m)}$. Any valid $(\Delta K, \Delta V)$ must therefore solve a system whose data is the set of all attractor positions. □ □

Remark 5 (Relation to Quantum Error Correction)**.** *System* ($\star$) *is structurally analogous to the* Knill–Laflamme conditions *for quantum error correction:*

$$P_C\, E_j^\dagger E_k\, P_C = c_{jk}\, P_C,$$

where correcting errors E_j against a codespace P_C requires knowing P_C. Here the codespace corresponds to the set of existing attractors, and the "error" is the global softmax weight shift induced by ΔK. Just as a quantum error correction protocol cannot be designed without knowing the code, an attention parameter update cannot preserve existing semantic attractors without knowing their positions. The implicit connectome plays the role of the quantum code: hidden, global, and required for any safe operation.

Conjecture 1 (Impossibility for Transformer Attractors)**.** *Let c_A, c_B be semantic attractors with $\langle c_A, c_B\rangle \approx 0$. Let a^* be a new concept such that c_{a^*} does not exist as a stable fixed point of the current attention iteration.*

Then any parameter change $\Delta\theta = (\Delta K, \Delta V)$ that introduces c_{a^*} as a new stable attractor necessarily disrupts at least one existing attractor, unless the full set of existing attractor positions $\{c_C^{(m)}\}$ is known and used to construct $\Delta\theta$.

Evidence*: By Lemma 8, safe preservation of existing attractors requires solving system* ($\star$)*, which depends explicitly on all attractor positions. Without this information, no ΔV can be guaranteed to satisfy the preservation condition. The remaining formal step—showing that a random or gradient-based $\Delta\theta$ without attractor knowledge fails with probability one—reduces to showing that a random vector generically fails to solve* ($\star$)*, which follows from the fact that* ($\star$) *is an overdetermined system for $p > d_v$ (more attractors than value dimensions), and from measure-theoretic arguments for $p \le d_v$.*

6 Consequences

6.1 Catastrophic Forgetting as Structural Necessity

The standard framing of catastrophic forgetting is algorithmic: the network "forgets" because gradient descent overwrites weights. Theorem 2 and Conjecture 1 reframe this:

For structurally new concepts, forgetting is not caused by a poorly chosen optimiser. It is caused by the impossibility of introducing a new eigenspace direction without globally reorganising the existing eigenstructure. The weight matrix W encodes semantic structure implicitly and globally: there is no submatrix of W that corresponds to a single concept. A new concept requires a new eigenvector, which requires shifting all others.

This explains why methods such as EWC, rehearsal, and LoRA succeed for new facts about existing concepts (Case 1 of Corollary 4) but fail for structurally new concepts (Case 2): they address the algorithmic symptom, not the structural cause.

6.2 Relation to Explicit Topology

The biological connectome avoids this limitation because its Laplacian L is sparse: a new synapse (i, j) modifies only four entries of L. By spectral perturbation theory, attractors spatially distant from (i, j) are affected only weakly. New topological paths arise locally without globally destabilising existing structure.

In contrast, a Transformer weight matrix W is dense: every gradient step shifts all singular values simultaneously, because what is local in the implicit semantic graph is global in W. This is the fundamental architectural difference between biological and artificial associative memory, and it explains why catastrophic forgetting is not an engineering deficiency but an inevitable consequence of implicit topology.

6.3 Implications for Architecture Design

Theorem 2 suggests that truly incremental concept learning requires architectures with explicit, locally modifiable topology—structures where adding a new concept corresponds to adding a local subgraph, not reorganising a global weight matrix. The biological connectome is one such architecture. Designing artificial analogues is an open engineering problem.

Proposition 9 (Minimal Architecture for Local Concept Addition)**.** *Let c_C be a semantic attractor and a^* a new concept to be added. By Lemma 7, the displacement of c_C is $g(c_C) = \alpha'_{a^*}(c_C)(v_{a^*} - c_C)$. The displacement vanishes if and only if $\alpha'_{a^*}(c_C) = 0$. Since $\alpha'_{a^*}(c_C) = \exp(c_C k_{a^*}^\top/\sqrt{d}) \,/\, Z > 0$ for any finite k_{a^*}, exact zero is unachievable. However, $\alpha'_{a^*}(c_C)$ can be made arbitrarily small by ensuring $c_C \cdot k_{a^*} \ll \max_j c_C \cdot k_j$ (the new key scores far below all existing keys for every existing attractor).*

Note: key orthogonality ($c_C \cdot k_{a^*} = 0$) is not *sufficient, since it yields $\alpha'_{a^*}(c_C) = e^0/Z = 1/Z$, which is only small when attention is already sharply peaked on existing keys. The correct sufficient condition requires negative alignment, not merely orthogonality. A practical construction is:*

$$k_{a^*} = -\, s\,(c_{a^*} - P_p\, c_{a^*}), \quad s \gg 1,$$

where $P_p = \sum_{i=1}^{p} e_i e_i^\top$ is the projection onto the existing attractor subspace and s is a scale factor chosen so that $c_C \cdot k_{a^*} \ll 0$ for all existing attractors c_C. In practice, gated routing (condition 3 below) achieves the same effect by zeroing the contribution of a^*'s head for non-matching inputs.

The minimal modification enabling local, non-destructive concept addition therefore consists of three conditions:

1. ***Appended weights**: new key-value rows (k_{a^*}, v_{a^*}) are added without modifying existing rows of K and V;*
2. ***Low cross-attention**: $c_C \cdot k_{a^*} \ll \max_j c_C \cdot k_j$ for all existing attractors c_C (negatively aligned keys, not merely orthogonal);*
3. ***Gated activation**: a routing mechanism ensures a^*'s attention head is active only for inputs near c_{a^*}, suppressing α'_{a^*} elsewhere.*

Conditions 2 and 3 jointly drive $\alpha'_{a^*}(c_C) \approx 0$, guaranteeing near-zero displacement of existing attractors. Mixture-of-experts with concept-specific gating satisfies all three.

7 Open Problems

1. **Quantitative forgetting bound**: *Resolved.* See Proposition 5: attractor ξ^m can survive the introduction of new concept a^* with target eigenvalue μ only if $|\mu - \lambda_m| < \delta_m$ (spectral gap condition). Any perturbation violating this condition has $\|\Delta W\|_2 \geq \delta_m$, beyond the Davis-Kahan safe radius.
2. **Formal proof of Conjecture 1**: *Substantially resolved* by Lemma 8 (Connectome Equivalence). Any parameter change $(\Delta K, \Delta V)$ that preserves all existing attractors while introducing a^* must solve system $(\star)$, whose data are the attractor positions themselves. Safe incremental learning is therefore equivalent to knowing the implicit connectome. The appended-key case is fully quantified by Lemma 7 and Remark 3. The remaining open step

is a measure-theoretic argument showing that gradient-based updates without attractor knowledge fail to solve ($\star$) with probability one.

3. **Threshold for Case 1 vs. Case 2**: *Resolved.* See Proposition 6: the operational threshold is $\cos\theta(a^*) = 1/\sqrt{p}$, where $\cos\theta = \|P_p a^*\|/\|a^*\|$ is the novelty angle.

4. **Architectural implications**: *Resolved.* Proposition 9 identifies the minimal modification: appended key-value pairs with concept-orthogonal keys and gated routing. Any architecture satisfying these three conditions admits incremental concept addition without catastrophic forgetting.

8 Conclusion

We have proved that in dense associative memory networks, no weight perturbation can introduce a structurally new concept without shifting at least one existing eigenvalue. This result is structural, not algorithmic: it holds for any symmetric ΔW of any rank, and no optimisation method can circumvent it.

The result establishes a categorical distinction between two learning cases: new facts about existing concepts (tractable in principle) and structurally new concepts (requiring global reorganisation). Catastrophic forgetting, in its most fundamental form, belongs to the second case.

For Transformer architectures, all four open problems are now substantially resolved. Three are fully proved: the attractor survival condition (Proposition 5), the geometric Case 1/Case 2 threshold (Proposition 6), and the minimal architecture for non-destructive concept addition (Proposition 9). The fourth—formally proving Conjecture 1—is resolved in a strong sense by the Connectome Equivalence Lemma (Lemma 8): any $\Delta(K, V)$ preserving existing attractors while introducing a^* must solve a linear system whose data are the attractor positions themselves, establishing that safe incremental learning is structurally equivalent to knowing the implicit connectome. The remaining open step is a measure-theoretic argument (probability-one failure without attractor knowledge).

The broader implication is architectural: truly incremental concept learning requires explicit, locally modifiable topology—specifically, appended key-value weights with concept-orthogonal keys and gated routing. Solving the problem of catastrophic forgetting for structurally new concepts is not a matter of better algorithms. It requires a different kind of machine.

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